

Master's Research

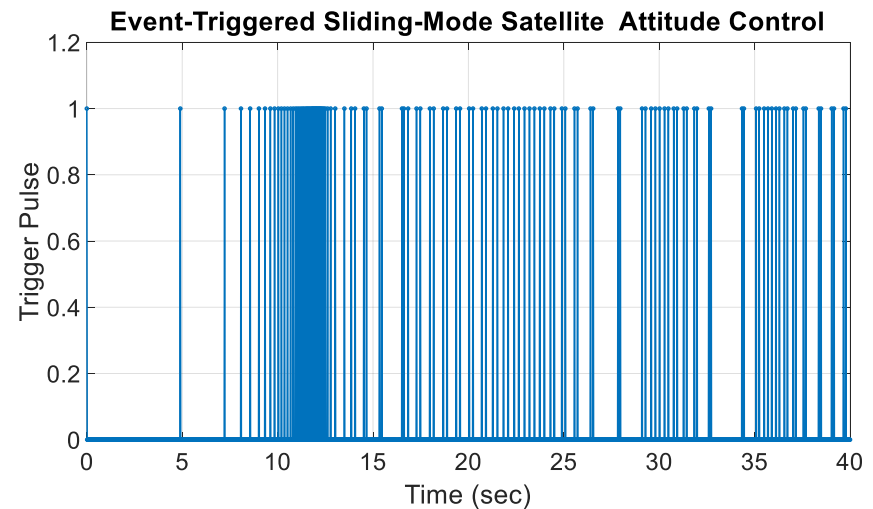
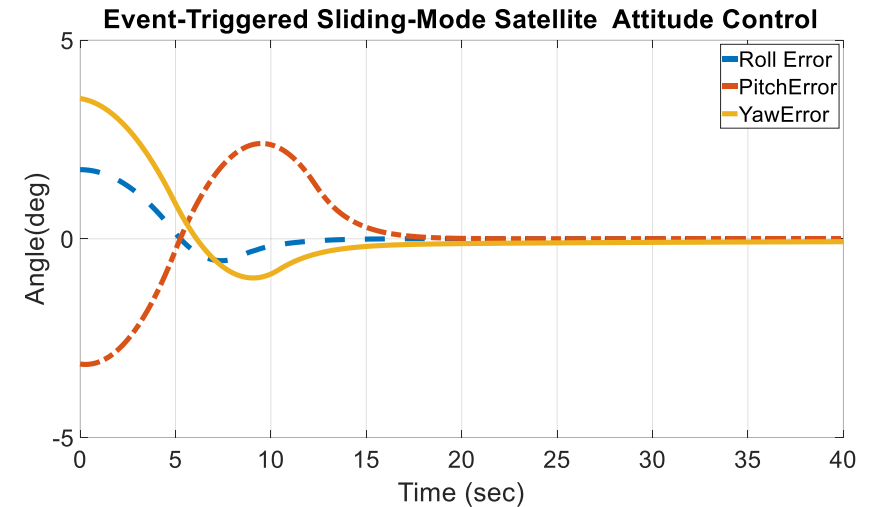
Advanced Satellite Attitude Control

1. Dynamics Modeling

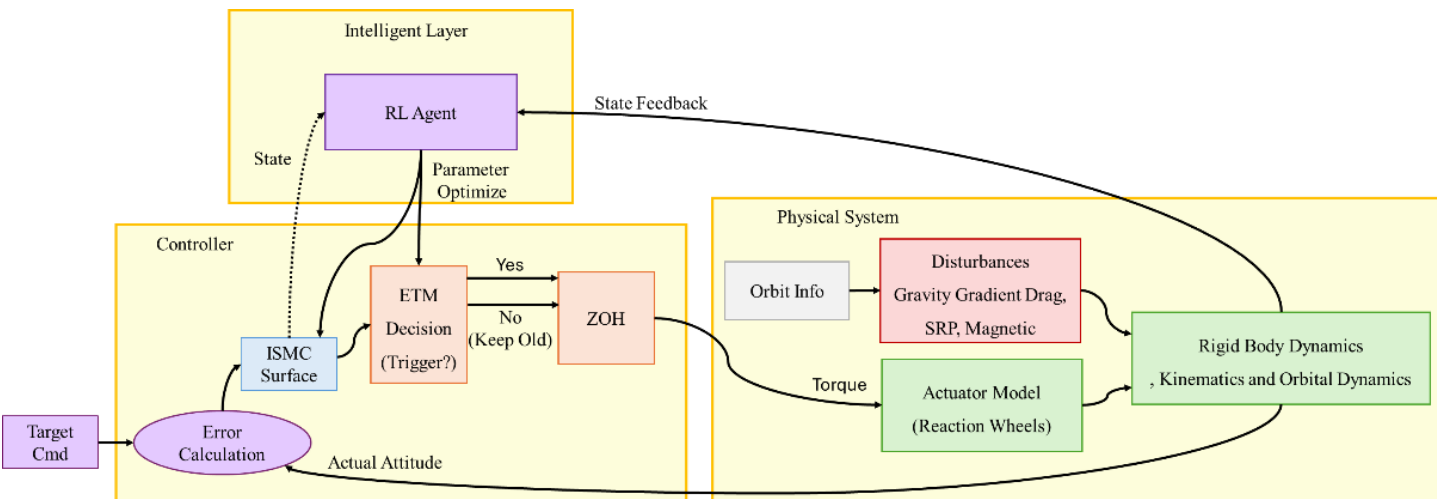
- $SO(3)$ Manifold & Newton-Euler Equations
- Disturbances: Gravity Gradient, SRP, and Magnetic torques
- Actuators: Reaction Wheel (Saturation/Friction)

2. Control Strategy

- ET-ISM: Event-Triggered + Integral Sliding Mode
- Intelligent Tuning: DDPG Reinforcement Learning
- Environment: MATLAB / Simulink



92.84% saving in actuator update



CubeSat Project

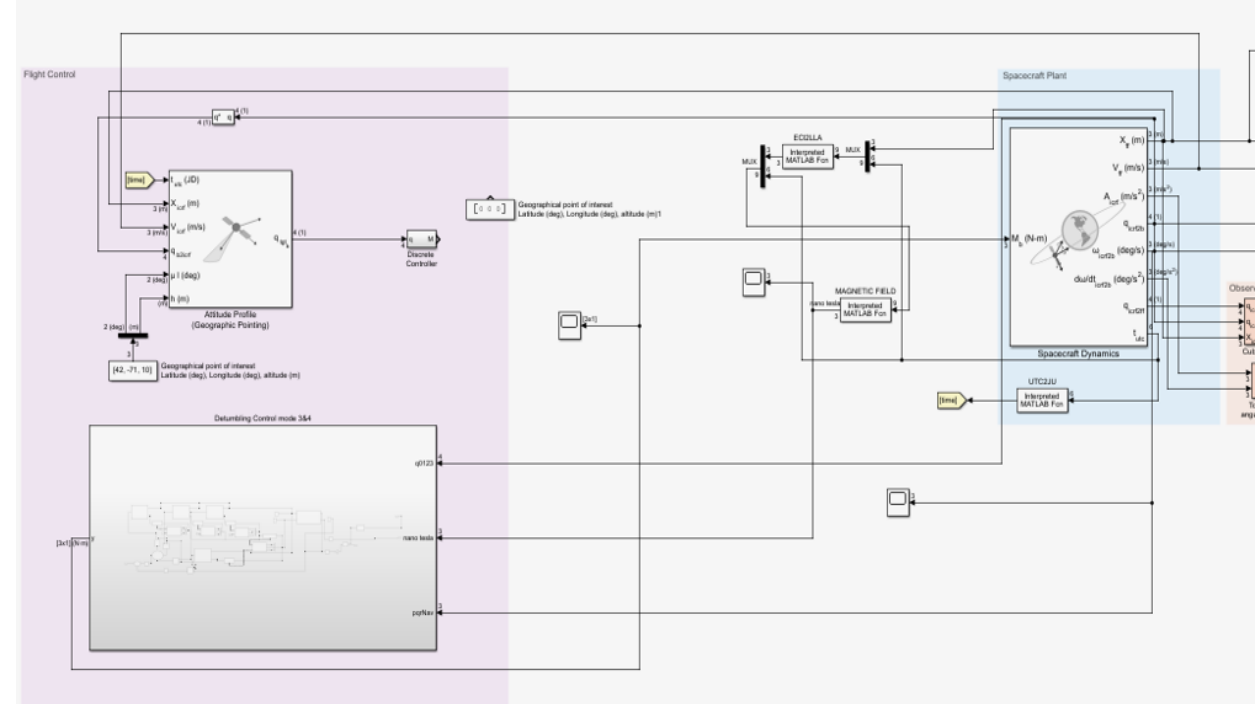
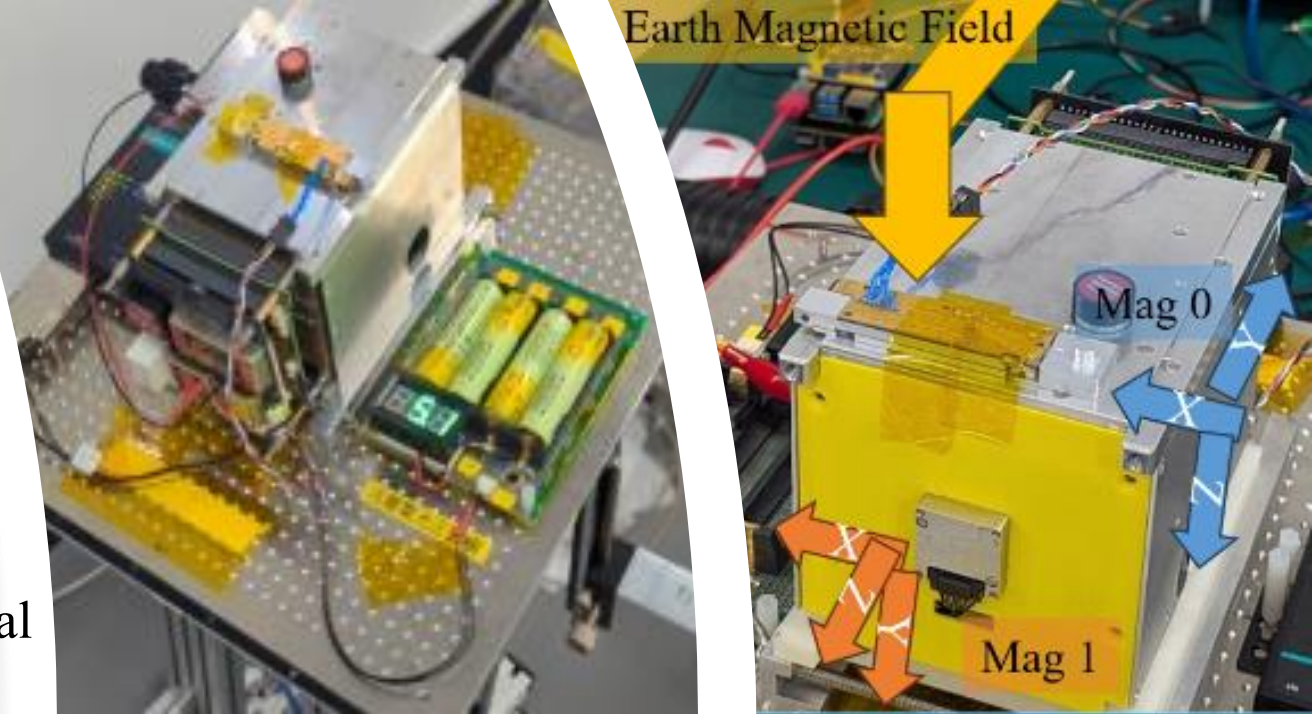
ADCS Subsystem

1. Hardware Integration

- Functional testing and firmware updates for external ADCS modules
- Component Verification: RW, Sun Sensors, and Magnetometers
- Defined and aligned Body Frame coordinates for all sensors/actuators

2. Control Validation

- Executed attitude control test on an air bearing table
- Verified control laws via hardware-in-the-loop
- Interface and communication debugging between OBC and ADCS



CubeSat Project

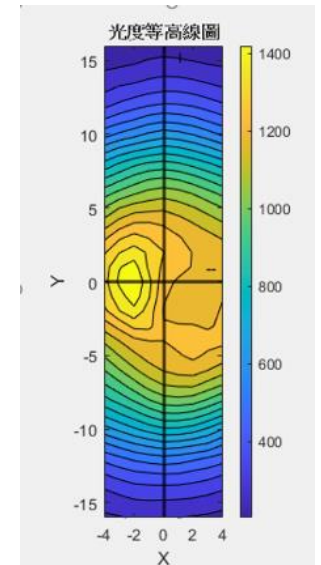
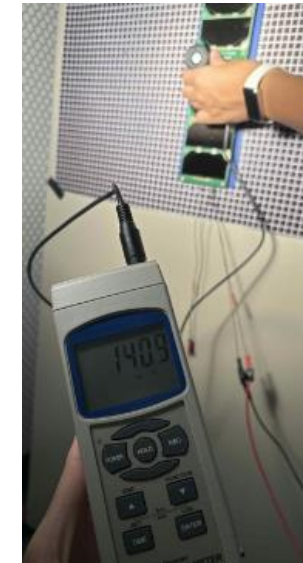
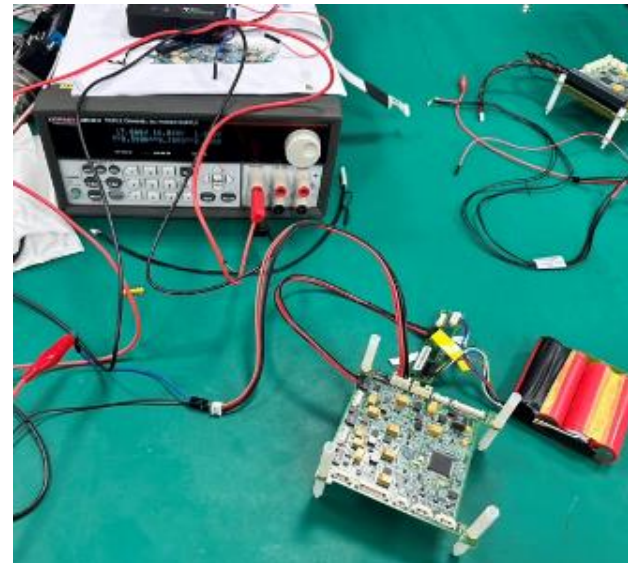
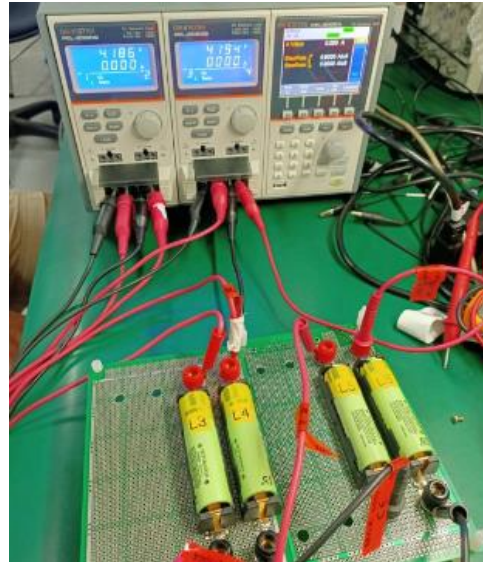
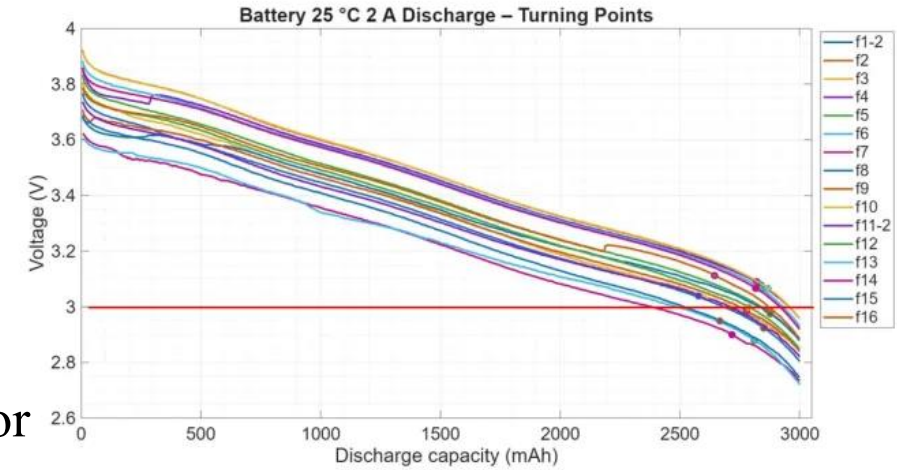
EPS Subsystem Development

1. Power Characterization and Verification

- Conducted discharge tests to map voltage-capacity curves
- Verified battery performance in Thermal Chambers
- Measured and analyzed solar panel conversion rates

2. System Integration and Debugging

- Resolved hardware-software interface issues with the EPS vendor
- Debugged communication protocols and telemetry data exchange



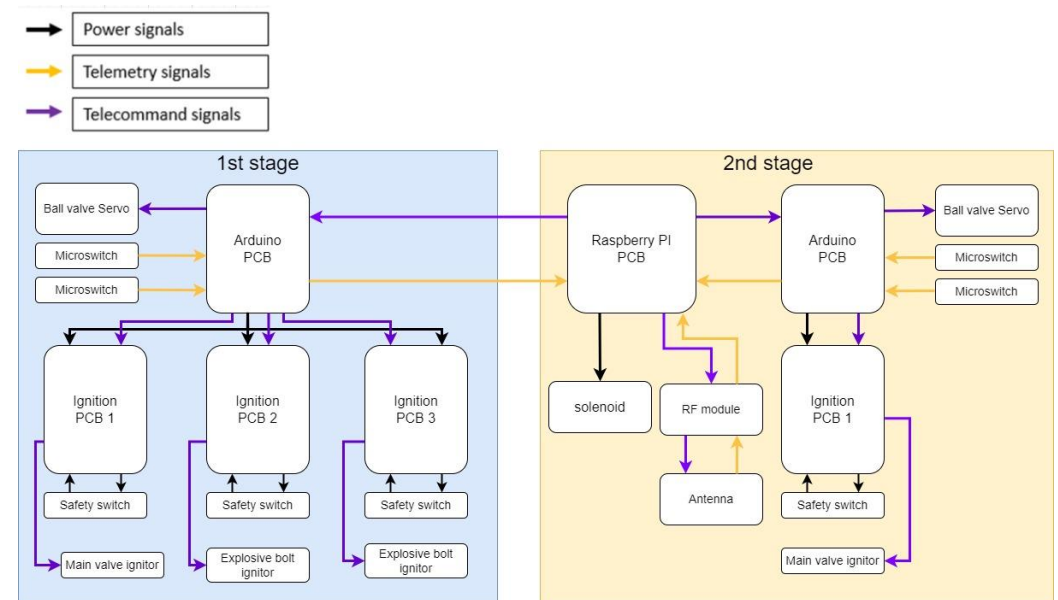
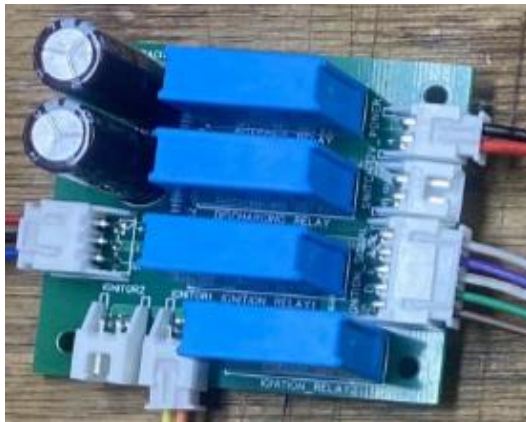
Capstone Project on Rocket ROS-Based Avionics & System Integration

1. Distributed Communication (ROS Bridge)

- Connected Raspberry Pi (high-level) and Arduino (low-level) via serial
- Published real-time data from pressure sensors and limit switches; subscribed to control commands
- Implemented C++ firmware for sensor acquisition and actuator driving

2. Circuit Development & System Integration

- Developed custom boards for Ignition/Relay Control and Sensor Power Conditioning
- Integrated distributed flight computers and custom circuitry into the mechanical structure



CanSat Competition

Avionics Team Leadership & Achievement

1. Avionics & Sensor Fusion

- Teensy 4.1 for high-speed multi-threaded mission control
- Integrated 10+ sensors via I2C, UART, and SPI (GPS, IMU, Pitot, Barometer)
- Developed real-time data logging and autonomous mission logic

2. Hardware & Verification

- Developed custom circular PCB with power regulation and RTC
- Validated full mission cycle via flight test, including autonomous deployment and real-time GPS tracking

